

Ekansh Chauhan

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EDUCATION

PhD in Cancer Sciences (Artificial Intelligence) , Cancer Research UK & Uni. of Glasgow, Scotland, UK	Exp. 2028
Master of Computer Science (Artificial Intelligence) , IIIT Hyderabad, India	2022-2024
Bachelor of Computer Science , GGSIPU University, Delhi, India	2017 - 2021

SKILLS

Languages:	Python, Java, SQL
<u>ML & LLM:</u>	AI Agents, LangChain/LangGraph, Foundation Models, Transformers, PyTorch
Systems & Deployment:	MLflow/LangFuse, PostGres/Redis/Minio, Linux, Docker, Git, Slurm
<u>Databases & APIs:</u>	MySQL, REST APIs, Vector Databases

PROJECTS

Multi-user self-hosted Biomedical AI Agent	Feb 2026 - Jul 2026
Cancer Research UK, Scotland Institute	<i>Glasgow, UK</i>

- Built a secure, multi-user biomedical AI research agent, deployed on institutional HPC, integrating LLM reasoning with Python, R, bioinformatics tools, and scientific databases.
- Automated end-to-end biomedical research workflows, reducing a representative ~2.5-week research task to ~11 minutes through autonomous literature synthesis, analysis, code execution, and hypothesis generation.
- Secured autonomous agent execution using Docker and Bubblewrap sandboxing, reverse proxy, and Langfuse observability.

Smart Lightweight Medical Query System	Aug 2024
Megathon'24	<i>Hyderabad, India</i>

- Built an edge-deployable medical QA system using LangChain + FAISS; reduced model size from 13.5 GB → 2.5 GB via distillation and quantization. [\[Link\]](#)

EXPERIENCE

Research Fellow - Computational Biology	Nov 2024 - Current
Cancer Research UK, Scotland Institute	<i>Glasgow, UK</i>

- Designed a multi-modal AI framework using Deep Learning and Sparse CCA to integrate high-dimensional single-cell transcriptomics with fine-scale tissue morphology.
- Validated framework on pan-cancer 10x Xenium datasets, identifying statistically significant gene-histology correlations.

Research Fellow - Cancer Diagnostics	May 2021 - Oct 2024
Centre for Visual Information Technology (CVIT), IIIT-H	<i>Hyderabad, India</i>

- Developed interpretable high-performance DL pipelines for giga-pixel histopathology imagery, achieving an 83.52% F1 score in breast cancer subtype classification and outperforming benchmarks by 5.35%.
- Curated a large-scale data lake of histopathology samples across 7+ cancer subtypes (Brain, Lung, Oral, etc.).

Computer Vision Intern	Dec 2023 - Feb 2024
Game Theory, Bangalore	<i>Bangalore, India</i>

- Deployed real-time, edge-optimized football tracking on Raspberry Pi 5 and Jetson Nano using TensorRT.
- Deployed a cloud-based 3D shuttle tracking system for badminton using multi-camera triangulation and YOLOv8.

SELECTED PUBLICATIONS

- **Nature** (2025): *MAPK-driven epithelial cell plasticity drives colorectal cancer therapeutic resistance* [\[Paper\]](#)
- **Nature Scientific Data** (2024): *IPD-Brain: An Indian histopathology dataset for glioma subtype classification* [\[Paper\]](#)
- **MIDL** (2024): *Lupus Nephritis Classification with only Slide-Level Labels* [\[Paper\]](#)

Note: Full publication list available on [Google Scholar](#)

ACHIEVEMENTS

- Served as a Teaching Assistant for Advanced NLP and Cognitive Science.
- Awarded the **Cancer Research UK** Core Fellowship for PhD studies in Cancer Sciences.
- Granted a fully funded fellowship by the **Department of Science and Technology, Government of India** for a MS by Research in Artificial Intelligence.
- Secured **1st rank** at the school level and **4th rank** at the state level in the Mathematics Olympiad.